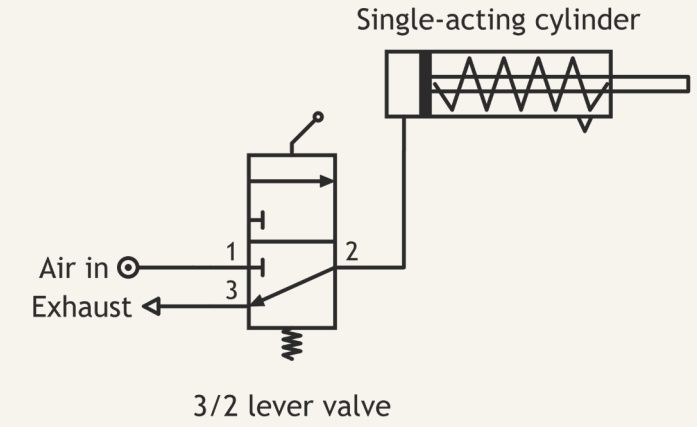
Daily Review

1	Regulator setting to change a tube?	0 bar
2	What pressure do we test at?	1 to 2 bar
3	Where must a circuit work first?	In the simulation
4	Three parts of a system?	Input, process, output
5	Energy of movement is called?	Kinetic energy

Reading the Diagram

Build the lever gate. Then draw it.



S3 Engineering Science

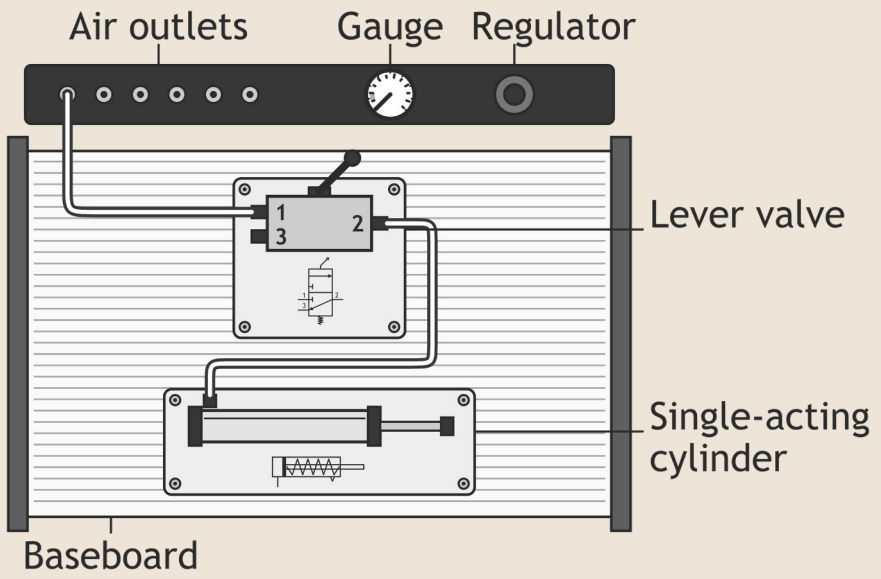
The Job: The Lever Gate

i When the lever is pushed, the rod moves out.

ii When the lever is let go, the rod moves back in.

iii The drawing shows every symbol and port.

Predict: you let go. Where does the air go?



By the end of this lesson…



1. I can say what a 3/2 valve does when it is pressed and let go
2. I can draw a circuit with symbols
3. I can build a circuit from a bench card

THE BUILD

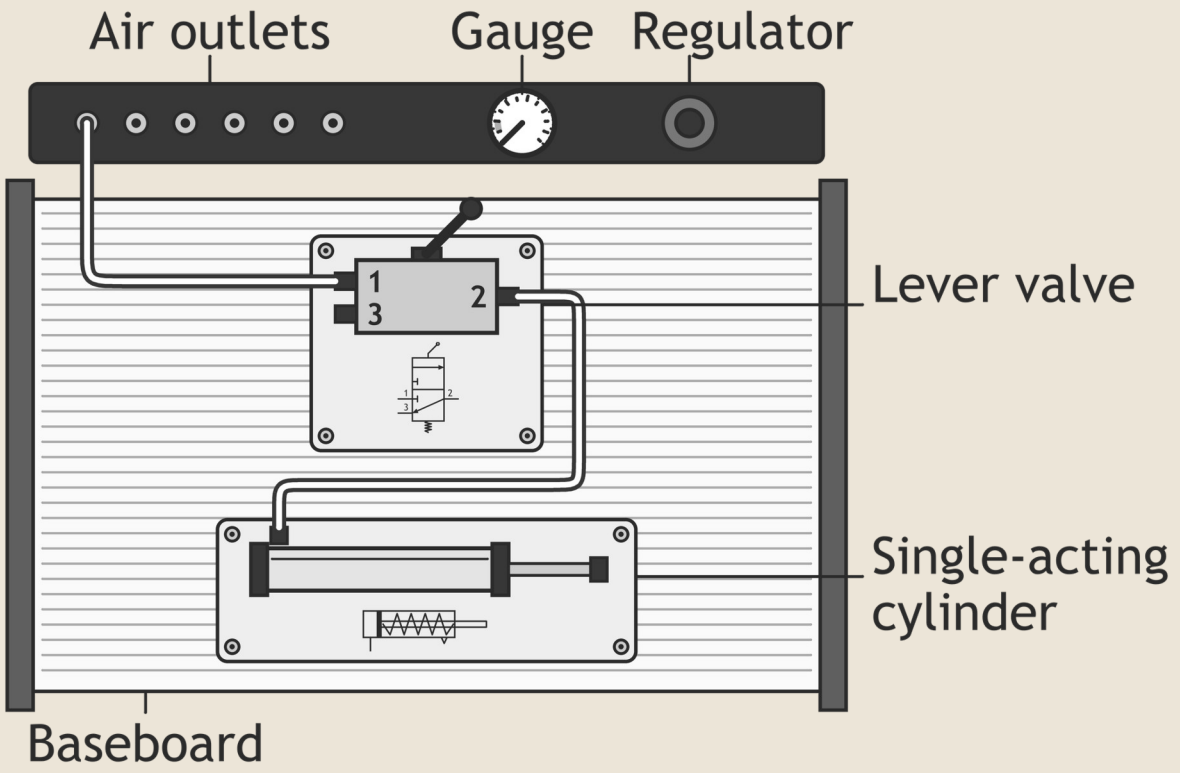
The Lever Gate

Simulate it. Air Permit. Then the kit.

What to Collect

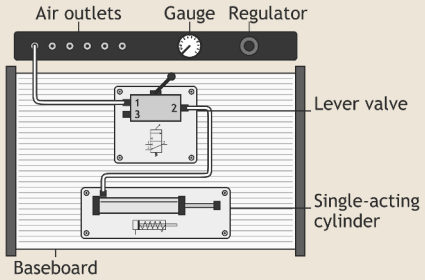
Builder and Safety: one set per team.

1 Bench card 2
2 Manifold (air supply)
3 3/2 lever valve
4 Single-acting cylinder
5 Tubing
6 Goggles, one each



Simulate and Build

Tasks 3-4 (B6A-L2-p6)

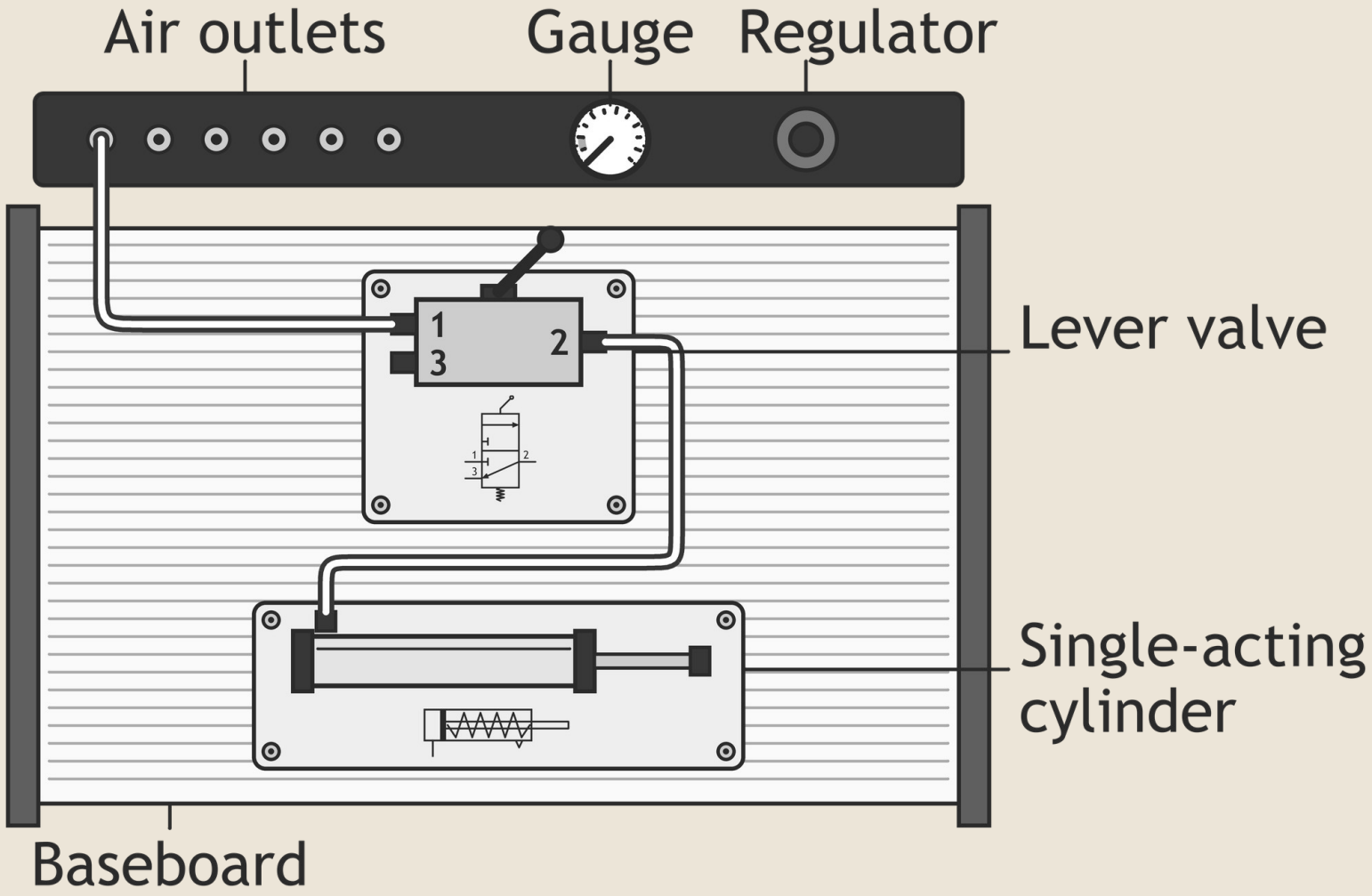


Tester and Recorder: start on the computer.
Builder and Safety: parts and bench card 2.
Air Permit checked? Push, let go, listen.

SAFETY Regulator at 0 to build. 1 to 2 bar to test.

10 min

Build It: Bench Card 2



Test It: What Did You Find?

1	Point i met? What pushed the rod out?	Yes: air from the valve
2	Point ii met? What pulled the rod back?	Yes: the cylinder's spring
3	Let go: where did the air go?	Out of the valve's exhaust

Parts of the Symbol (1 of 2): Boxes and Ports

Box

one state of the valve

2 boxes

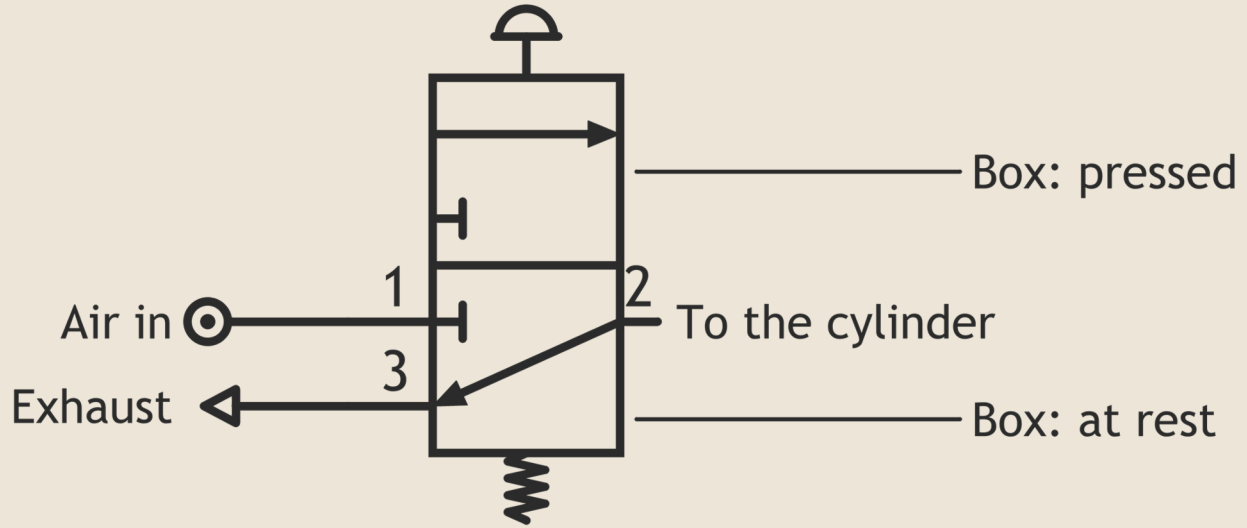
pressed and at rest

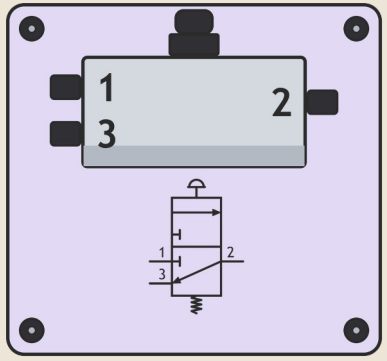
Ports

1 in, 2 out, 3 exhaust

3/2

3 ports and 2 states





The real valve. Its symbol is printed on the plate.

Parts of the Symbol (2 of 2): Inside the Boxes

Arrow

the path the air takes

T

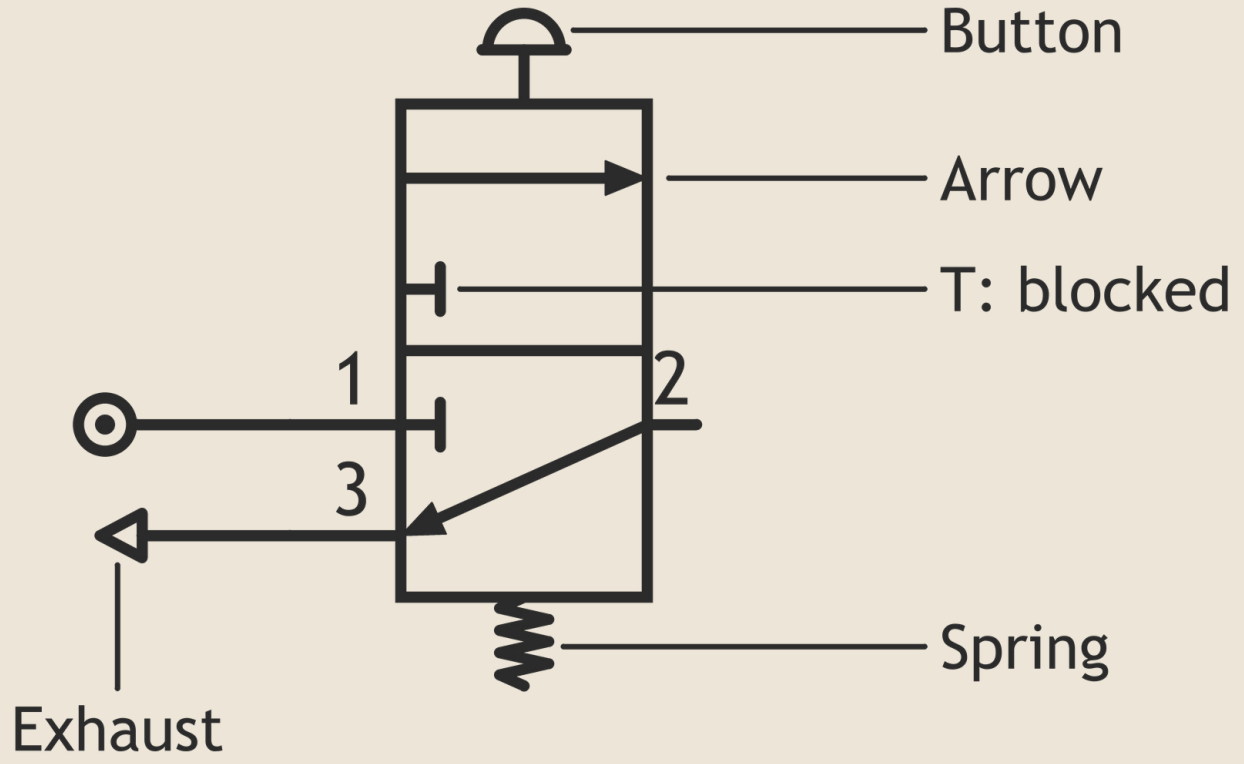
that port is blocked

Button

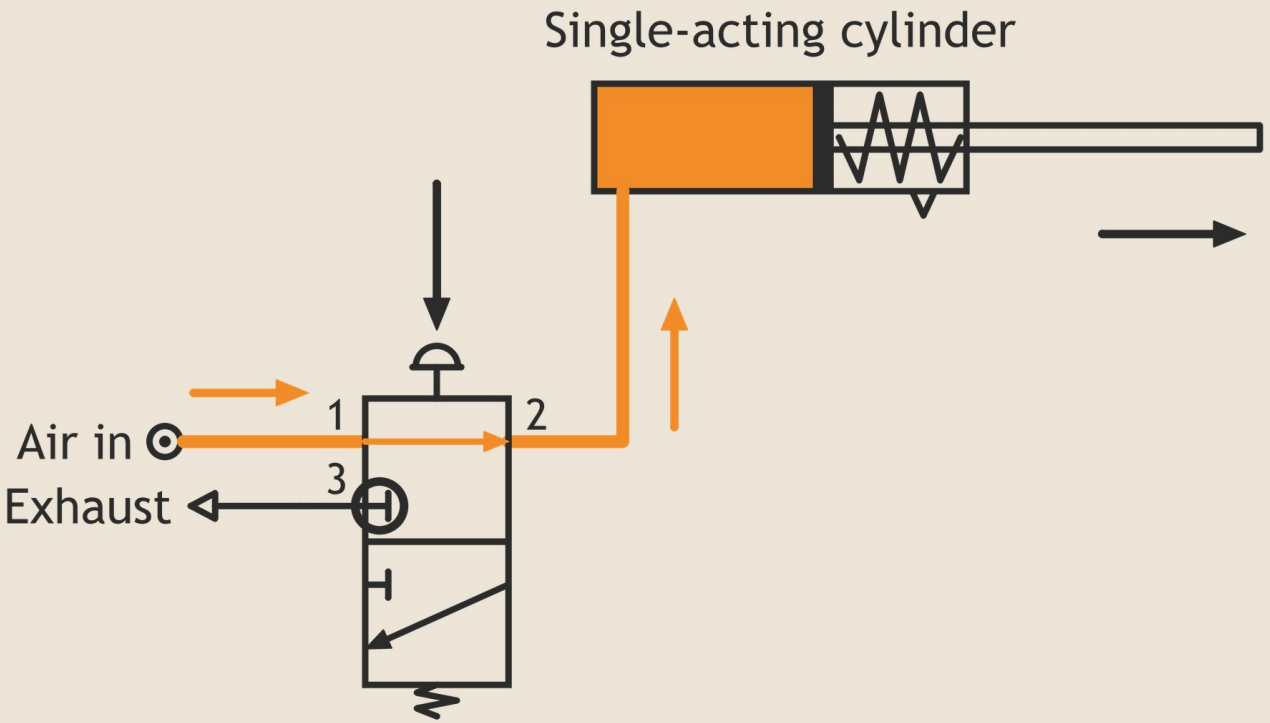
push it to change state

Spring

pushes the valve back



Worked Example: Press the Button



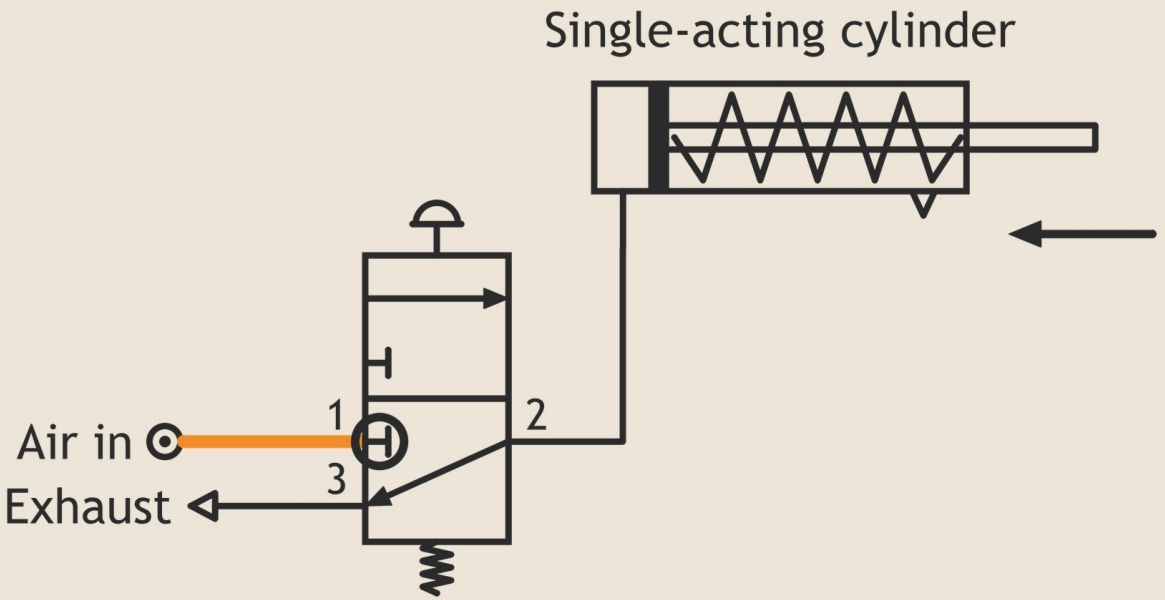
1 Push the button: both boxes slide down.

2 Port 3 is now blocked.

3 Air flows in port 1, out of port 2.

4 The rod moves out: the cylinder extends.

We Do: Let Go



1 Let go: the spring pushes the boxes up.

2 Port 1 is now blocked.

3 Air leaves: in port 2, out of port 3.

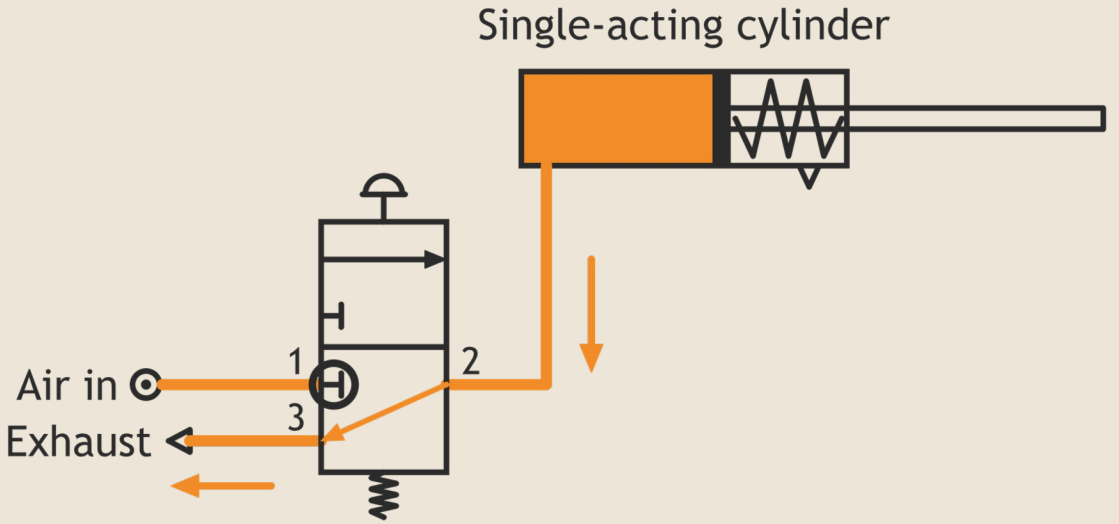
4 The rod moves in: the cylinder retracts.

Spot the Mistake

A pupil wrote: “When I let go, air goes into port 3 and pushes the rod back in.”

Find the mistake. Fix it.

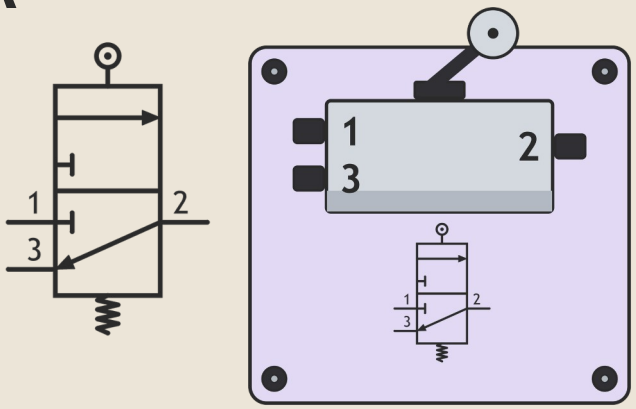
Air goes OUT of port 3. Port 1 is blocked. The spring pushes the rod in.



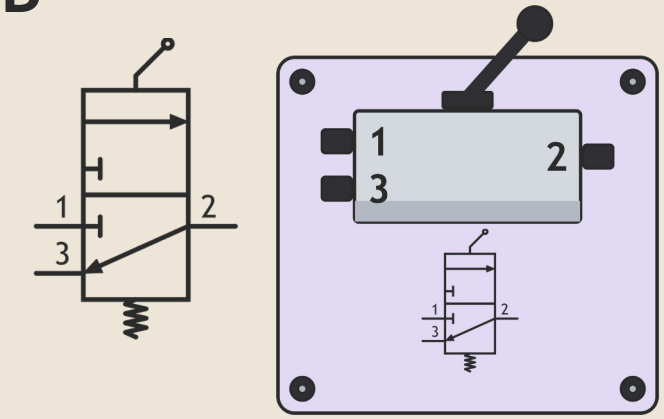
Show Me: Same Valve, Different Push

Same boxes, same ports. Name the part on top.

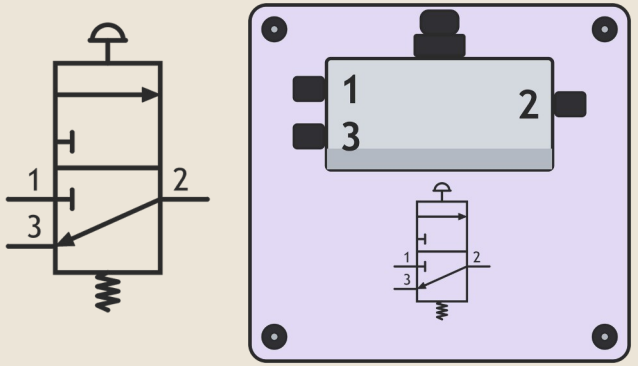
A



B



C



A Roller: a train presses it

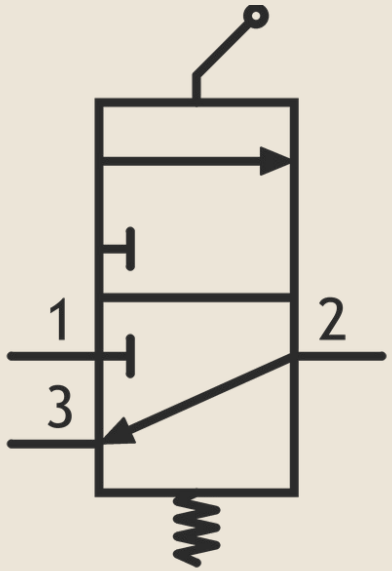
B Lever: a hand moves it

C Button: a finger pushes it

Now You, Then Your Turn

Task 1 (B6A-L2-p5)

Question 1: the lever valve, with a partner.
Question 2: the roller valve, on your own.
Use the Let go steps to help.



5 min

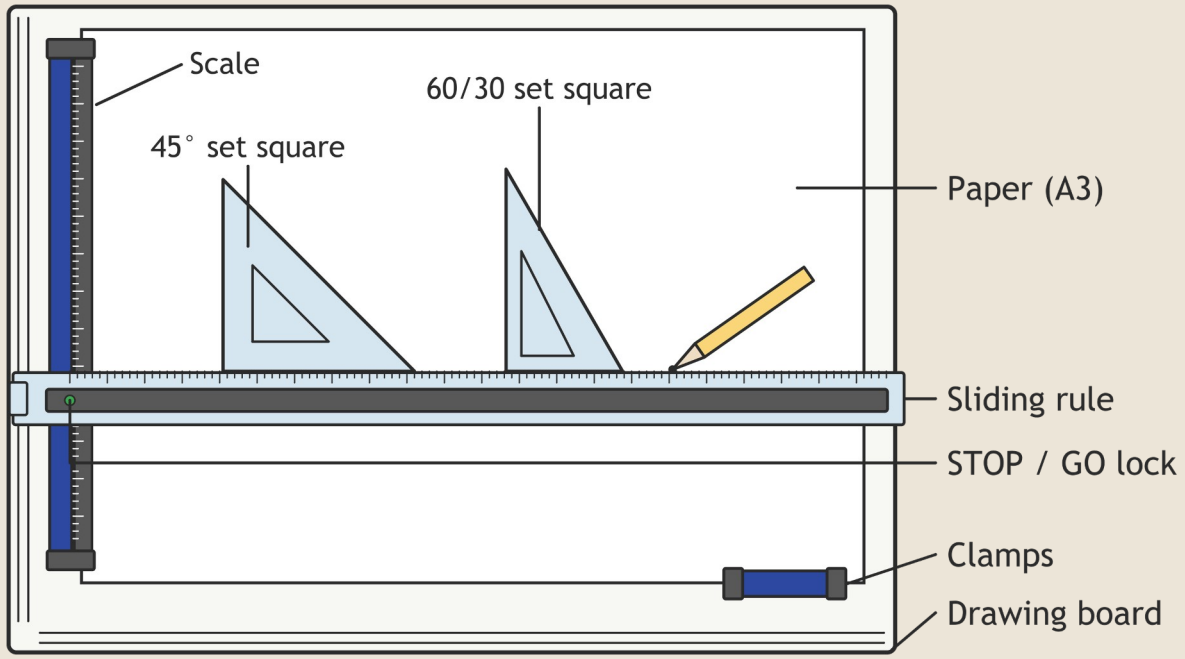
Answers: Task 1 (B6A-L2-p5)

Question	Answer
Q1 line 2	Into the cylinder: it pushes the piston
Q1 line 3	The rod moves out: the cylinder extends
Q2 mark 1	Roller pressed: the rod moves out
Q2 mark 2	Train gone: valve springs back, rod goes in

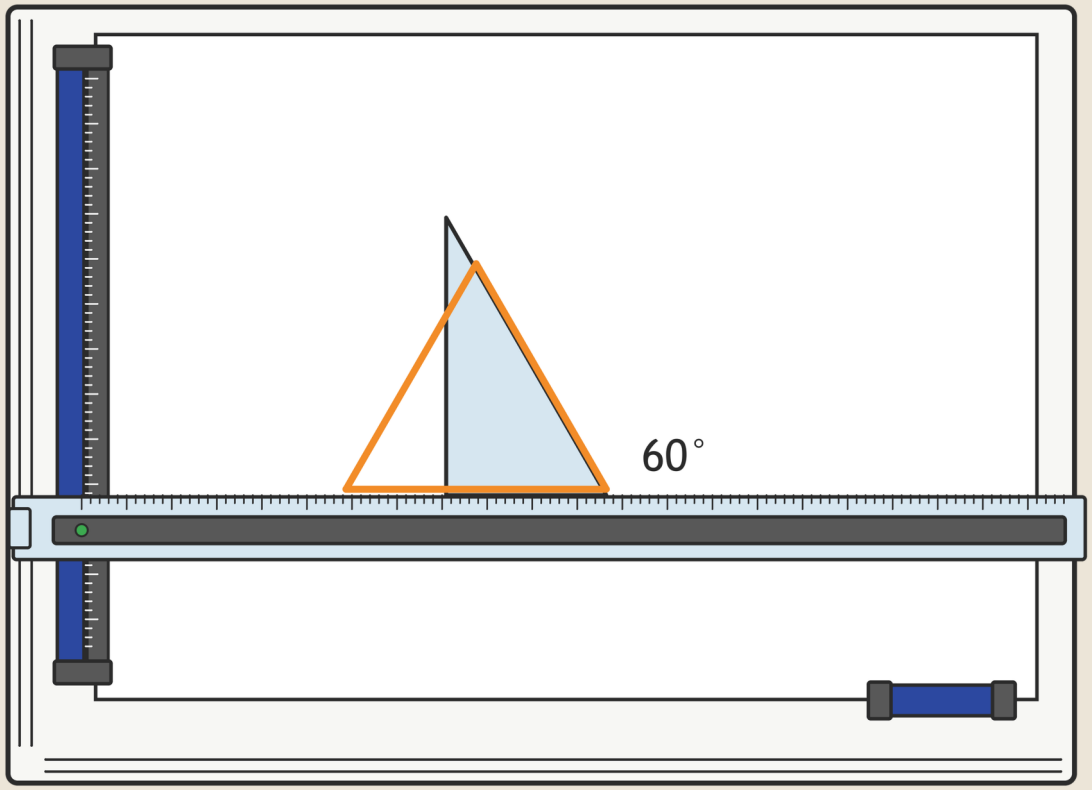
What to Collect

Collect one of each.

1 Drawing board
2 Plain A3 paper
3 45° set square
4 60/30 set square
5 Sharp pencil
6 Eraser



Using the Board



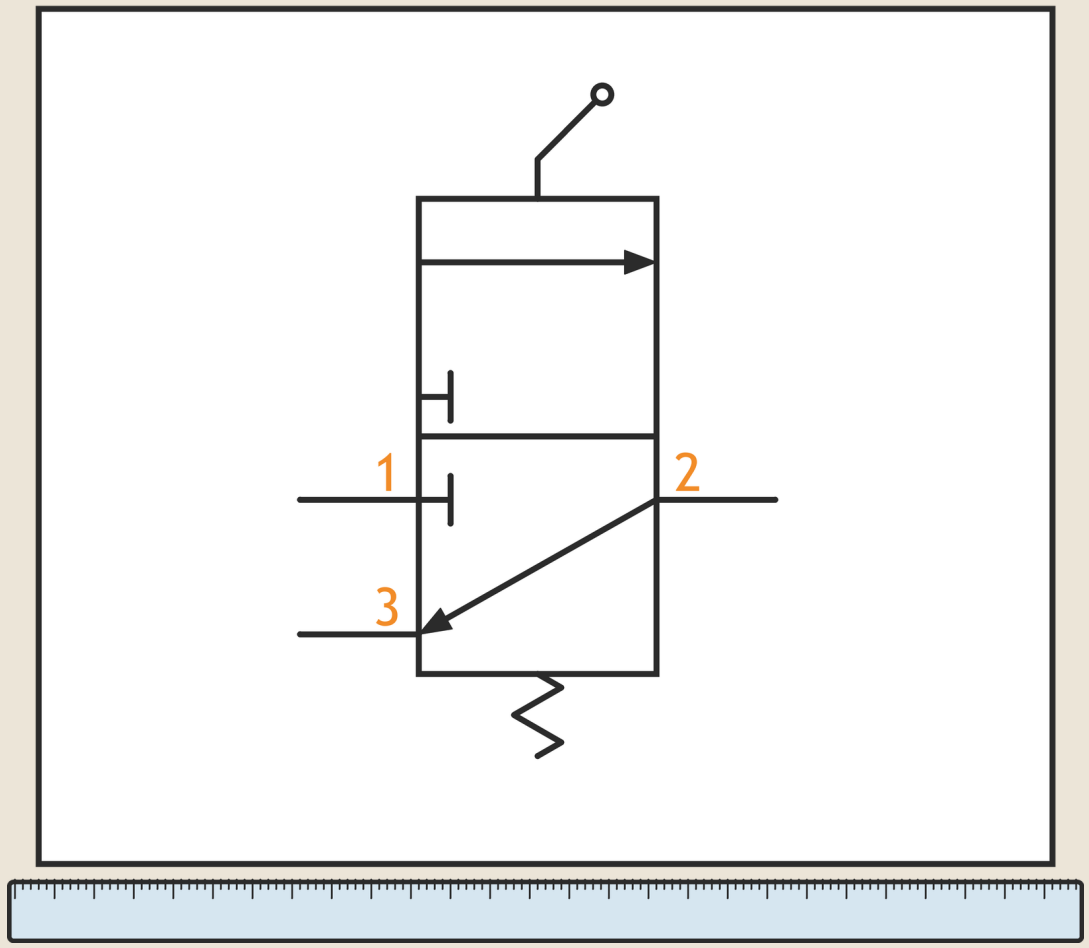
1 Clamp: paper in the bottom left corner.

2 Horizontal: pencil along the top of the rule.

3 Vertical: set square on the rule.

4 Angles: the 60° corner draws the exhaust.

Draw the Valve, Full Size



1 Two 30 mm squares, one on top.

2 Port lines, 15 mm long.

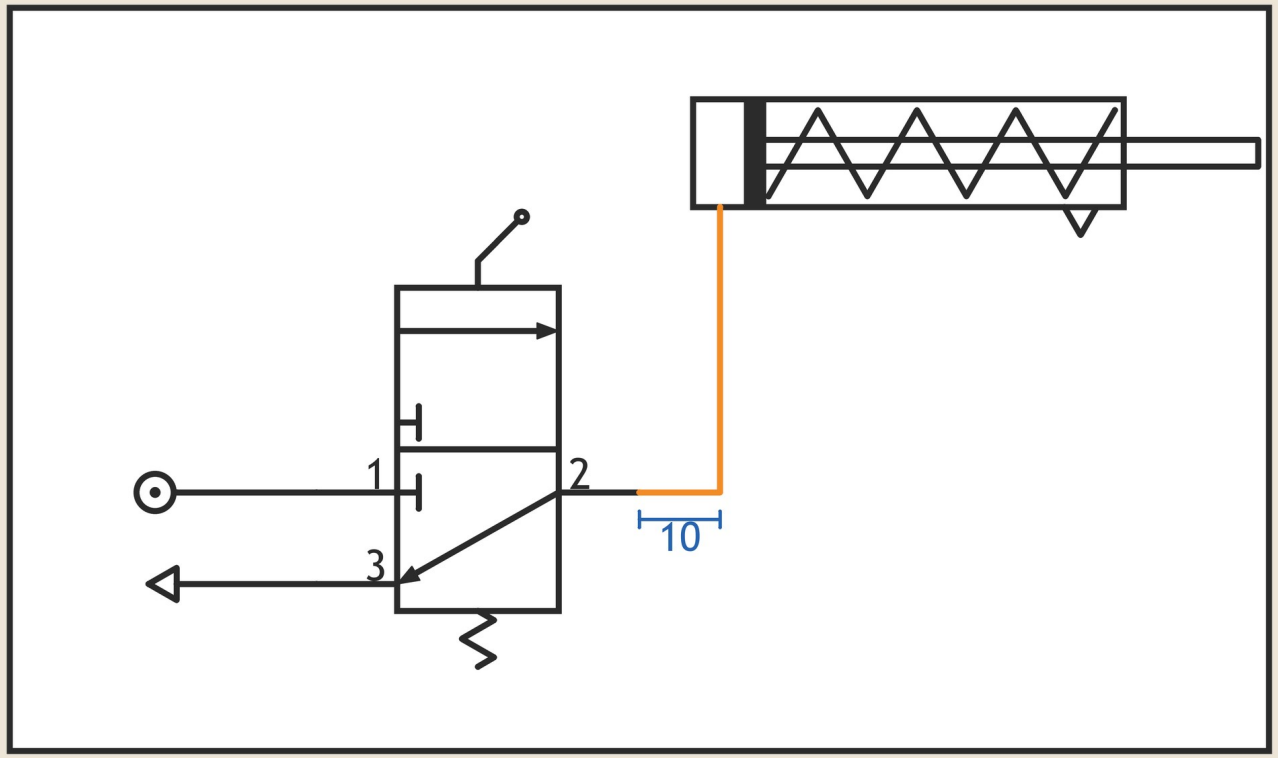
3 Straight arrow and two Ts.

4 Diagonal: 60/30 set square, 30° corner left.

5 Lever at 45°. Spring at 30°.

6 Number the ports 1, 2, 3.

Draw the Rest, Full Size



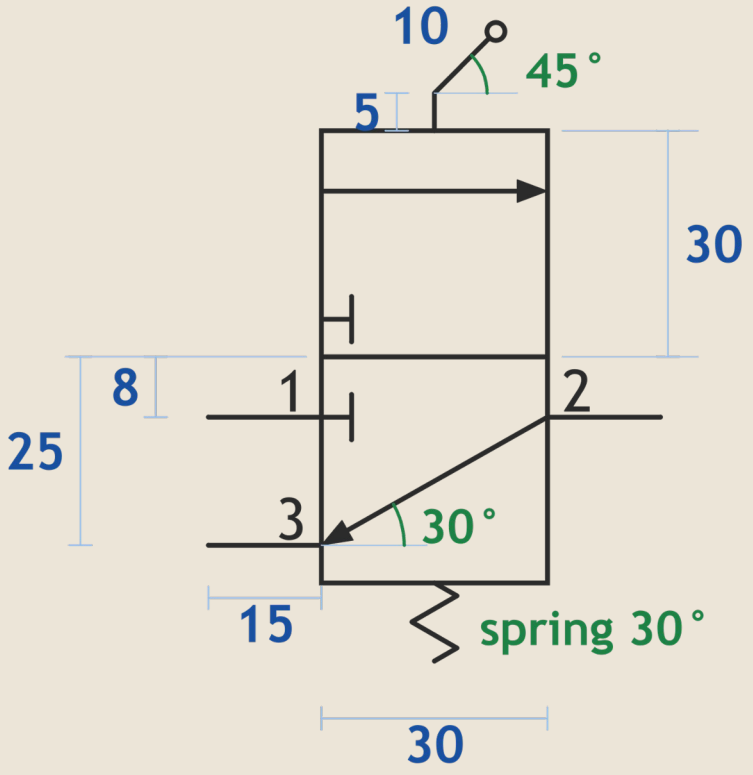
1 Cylinder 80 × 20 mm; rod 25 mm out.

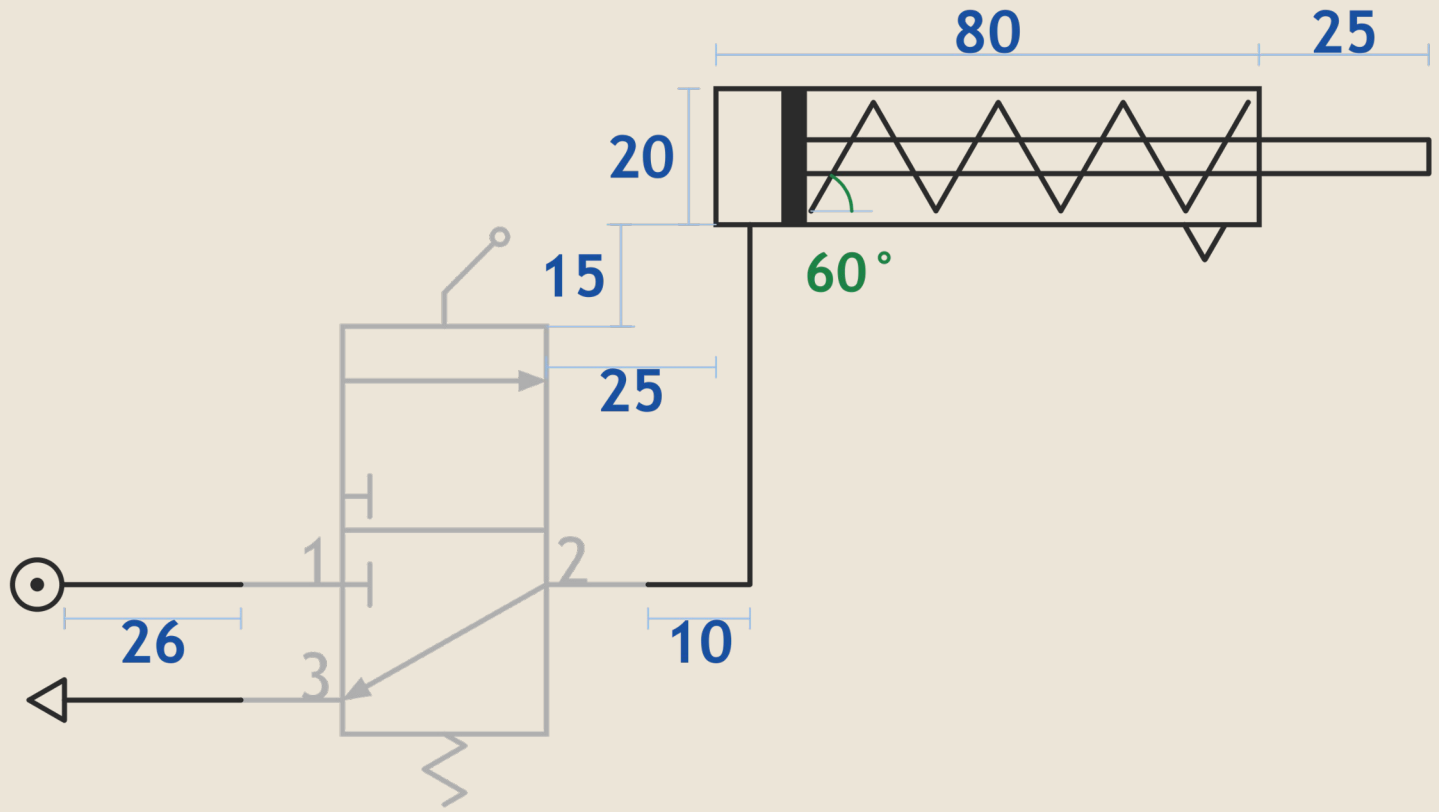
2 Spring at 60°; small triangle below.

3 Air in: 8 mm circle. Exhaust: 6 mm triangle.

4 Pipe: 10 mm across, then up.

Draw the Gate Circuit





Task 2 (B6A-L2-p5): plain A3, lever valve, these sizes.
Small parts by eye. Pipes across or up only.

7 min

Check Your Drawing

Sizes

the same as the guide

Ports

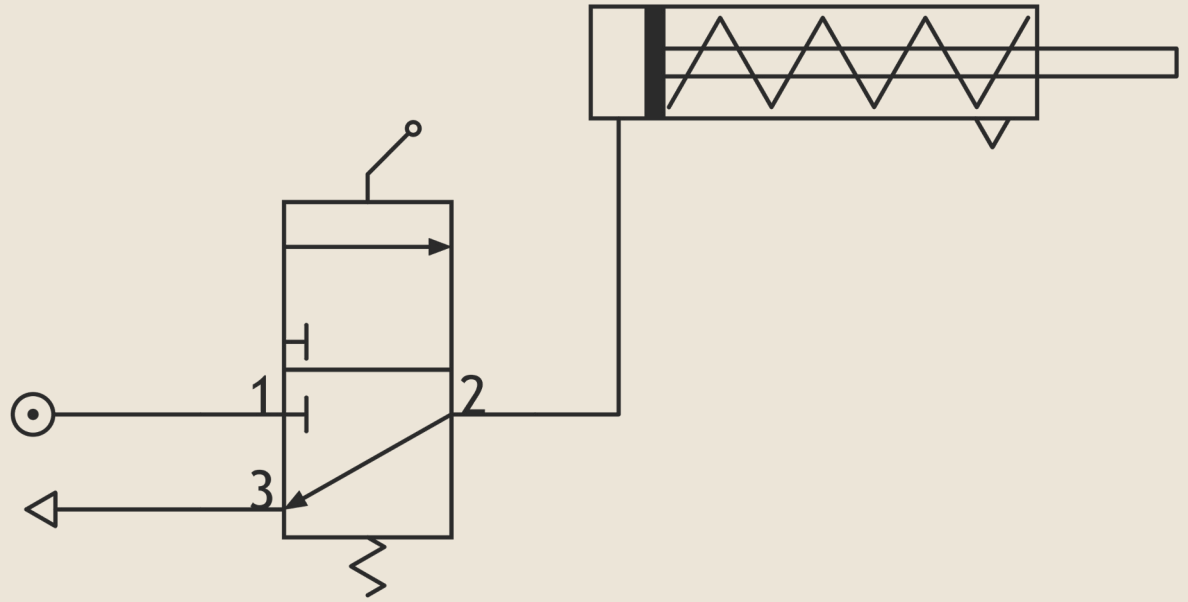
numbered 1, 2, 3

Exhaust

on port 3

Pipes

across or up only



Questions and Key Words

Tasks 5-6 (B6A-L2-p6)

Task 5: Questions 3 and 4 on your own.

Task 6: Key words.

4 min

Answers: Tasks 5-6 (B6A-L2-p6)

1	3 Why is it called a 3/2 valve?	3 ports and 2 states
2	4 The gauge before you change a circuit?	0 bar
3	Key words, line 1	ports, states
4	Key words, lines 2 and 3	1, 2; exhaust

Plenary: Reading the Diagram

1	How many states has a 3/2 valve?	2
2	What does a T in a box mean?	That port is blocked
3	Button pressed: which port is blocked?	Port 3
4	Let go: what pushes the valve back?	The spring
5	Which edge draws a horizontal line?	Top edge of the rule

Look back — by the end of this lesson…



1. I can say what a 3/2 valve does when it is pressed and let go
2. I can draw a circuit with symbols
3. I can build a circuit from a bench card

LOOKING AHEAD

Lesson 3: Control the Flow

How do you make the rod move slowly?

Credits

The job and the kit: Sources: job card and kit drawings by R Stewart.

Gate circuit: The gate circuit adapted from the Matrix CP2080 Pneumatics Essentials Worksheet 1 (© Matrix Technology Solutions Limited). Drawings by R Stewart.

Valve symbol and gate circuit pictures: The gate circuit adapted from the Matrix CP2080 Pneumatics Essentials Worksheet 1 (© Matrix Technology Solutions Limited). Drawings by R Stewart.

Real valve drawings: Sources: kit drawings by R Stewart, from the Matrix CP2080 kit.

What to Collect and Using the Board: Sources: drawing board pictures drawn by R Stewart from photos of the class boards.

R Stewart - October 2026